

Backend API Reference

All routes are mounted under the context path `/api/v1`. Controllers live in `Sample/backend/src/main/kotlin/com/cocoa/web/controller/`. “Required Authority” lists the Spring Security `@PreAuthorize` permission string; “none declared” means the endpoint is reachable by any authenticated user (protected only by the global JWT filter).

Authentication — `/auth`

Method	Path	Required Authority	Function
GET	<code>/auth/me</code>	<code>read:profile:own</code>	Returns the logged-in user's profile
POST	<code>/auth/register</code>	<code>public</code>	Self-registration for a new user
POST	<code>/auth/login</code>	<code>public</code>	Authenticates credentials, sets JWT as an HTTP cookie
GET	<code>/auth/logout</code>	<code>authenticated</code>	Clears the JWT cookie
PATCH	<code>/auth/reset-password</code>	<code>update:profile:own</code>	Updates password, clears the current session cookie

Admin — `/admin`

Method	Path	Required Authority	Function
POST	<code>/admin/users</code>	<code>create:user:any</code>	Creates a new researcher account with a forced password-reset flag

Researchers — `/researchers`

Method	Path	Required Authority	Function
POST	<code>/researchers/register</code>	<code>create:researcher:own</code>	Links a researcher profile (name, org) to the calling user
GET	<code>/researchers</code>	<code>read:researcher:all</code>	Lists/searches researchers with query filters

GET	/researchers/me	read:researcher:own	Returns the caller's own researcher profile + roles
GET	/researchers/{userId}	read:researcher:all	Gets a specific researcher by ID
PATCH	/researchers/{userId}	update:researcher:own	Updates a researcher profile (code also enforces caller can only edit self)
DELETE	/researchers/{userId}	delete:researcher:any	Deletes a researcher profile

Forms — /forms

Method	Path	Required Authority	Function
GET	/forms	read:form:all	Lists all available data-collection forms
GET	/forms/{formId}	read:form:all	Gets the structure/schema of one form
PUT	/forms/{formId}/edit	update:form:all	Updates a form's configuration

Tasks — /tasks

Method	Path	Required Authority	Function
GET	/tasks	none declared	Lists all tasks (form-filling assignments)
GET	/tasks/{taskId}	none declared	Gets a single task's details

Task Responses — /tasks/{taskId}/responses

Method	Path	Required Authority	Function
GET	/tasks/{taskId}/responses	none declared	Lists submitted form responses for a task
GET	/tasks/{taskId}/responses/{responseId}	none declared	Gets one specific form response

Analytics — /analytics

All endpoints require `read:report:all` and take a date-range filter (`from/to`, validated `from ≤ to`).

Method	Path	Required Authority	Function
GET	/analytics/harvest/time-series/sum	read:report:all	Cumulative harvest volume over time
GET	/analytics/harvest/time-series/delta	read:report:all	Month-over-month change in harvest volume
GET	/analytics/harvest/time-series/average	read:report:all	Average harvest amount per month
GET	/analytics/harvest/time-series/frequency	read:report:all	Count of harvest records per month
GET	/analytics/harvest/summary/total	read:report:all	Total harvest quantity in range
GET	/analytics/harvest/summary/count	read:report:all	Total number of harvest records in range
GET	/analytics/users/time-series/sum	read:report:all	Cumulative user growth over time
GET	/analytics/users/time-series/delta	read:report:all	Monthly change in user count
GET	/analytics/users/summary/count	read:report:all	Total user count in range

Spatial Analytics — /analytics/harvest/spatial

All require `read:report:all`. POST because each call carries a GeoJSON polygon body defining the map-drawn area, plus `from/to/gradeCode` query params.

Method	Path	Required Authority	Function
POST	/time-series/delta	read:report:all	Monthly harvest change within the drawn area
POST	/time-series/sum	read:report:all	Cumulative harvest within the drawn area
POST	/time-series/average	read:report:all	Monthly average harvest weight within the area

POST	/time-series/frequency	read:report:all	Monthly event count within the area
POST	/summary/total	read:report:all	Total harvest weight within the area
POST	/summary/count	read:report:all	Total event count within the area

Reports — /reports

Method	Path	Required Authority	Function
GET	/reports/raw-data/download?sheets=FARM,HARVEST	none declared	Generates and streams back an .xlsx file (Apache POI) with the requested raw-data sheets; defaults to all sheets if none specified

Test/Debug — (root)

Method	Path	Required Authority	Function
GET	/public/test	public	Sanity-check route, returns a static string
GET	/test	authenticated	Sanity-check route to verify JWT auth is working